The Fixed Bartlett Correction for Structural Equation Models

A Complete Derivation from General Theory

A Research Monograph Based on:

Cordeiro, G. M. & Cribari-Neto, F. (2014). *An Introduction to Bartlett Correction and Bias Reduction.* Springer Briefs in Statistics.

Savalei, V. (2010). *Small Sample Statistics for Incomplete Nonnormal Data.* Structural Equation Modeling, 17(2), 241–264.

May 21, 2026

Contents

1 Introduction and Historical Context

The likelihood ratio (LR) test statistic is one of the most widely used tools for testing hypotheses in parametric models. Under standard regularity conditions, the LR statistic converges in distribution to the χ^2_q distribution as the sample size $n \to \infty$, where q is the number of restrictions imposed by the null hypothesis. This first-order asymptotic result provides approximate critical values for conducting tests. However, when the sample size is small or moderate, these critical values can be inaccurate, leading to size distortions—that is, the effective Type I error rate may deviate substantially from the nominal significance level selected by the practitioner.

The fundamental problem is that the expectation of the LR statistic under the null hypothesis is not exactly q in finite samples. Rather, it takes the form

$$\mathbb{E}(w) = q + b + \mathcal{O}(n^{-2}), \tag{1}$$

where b is a constant of order $\mathcal{O}(n^{-1})$ that depends on the specific model structure and parameter configuration. This n^{-1} bias in the mean of w is the primary source of finite-sample size distortion.

The idea of correcting the LR statistic by dividing by a suitable factor to improve its χ^2 approximation was pioneered by Bartlett (1937) in the context of comparing variances of several populations. Bartlett observed that by introducing a correction factor $c = 1 + b/q$, the transformed statistic

$$w^* = \frac{w}{c} = \frac{w}{1 + b/q} \tag{2}$$

has its first three cumulants matching those of χ^2_q with error of order $\mathcal{O}(n^{-2})$, rather than the $\mathcal{O}(n^{-1})$ error that characterizes the uncorrected statistic. This remarkable result was later generalized by Lawley (1956), who derived a general formula for b as a function of covariant tensors formed from the joint cumulants of log-likelihood derivatives.

In the domain of structural equation modeling (SEM), the standard practice is to use the Satorra–Bentler scaled chi-square statistic T_{SC} to correct for nonnormality. However, T_{SC} itself requires medium to large samples to perform well. Various *ad hoc* small-sample corrections have been proposed, including the so-called "Bartlett-corrected scaled chi-square" of Savalei (2010, Eq. 7), which applies an adjustment factor derived from Bartlett's original 1950 factor analysis work. As Savalei herself acknowledges, this correction is ad hoc: its theoretical justification is lost when extended to general covariance structure models.

The purpose of this monograph is to provide a rigorous, step-by-step derivation of a **Fixed Bartlett Correction for SEM** that is grounded in the general Bartlett correction theory of Lawley (1956), Hayakawa (1977), and Cordeiro (1983, 1987). We begin with the original Bartlett (1950) formula, develop the full general theory through Bartlett identities and Lawley's expansion, and then apply this machinery to derive a properly justified correction for the SEM likelihood ratio test. We name this the **Cordeiro–Savalei Fixed Bartlett Correction** to distinguish it from the ad hoc variants currently in use.

2 The Standard Bartlett Correction (1950)

Definition 2.1 (Standard Bartlett Correction — Bartlett, 1950)**.** In the context of testing whether k principal components are sufficient to reproduce a $p \times p$ correlation matrix, the standard Bartlett correction adjusts the LR test statistic as:

$$T^*_{\mathrm{ML}} = \frac{T_{\mathrm{ML}}}{c}, \qquad c = 1 + \frac{2p + 4k + 5}{6(n-1)}, \tag{3}$$

where T_{ML} is the ML chi-square test statistic, p is the number of observed variables, k is the number of retained principal components (or latent factors), and n is the sample size.

Remark 2.1. This formula was originally derived by Bartlett (1950) specifically for the test of sphericity in principal component analysis. Its theoretical foundation rests on the special structure of the covariance matrix under the principal component model, which allows the cumulants of the log-likelihood derivatives to be computed in closed form.

The key insight is that under the principal component model, the n^{-1} correction to the expected value of T_{ML} takes a particularly simple form. The numerator $2p + 4k + 5$ arises from the specific way in which the parameter count and the dimensionality of the model interact through the log-likelihood derivatives. The denominator $6(n-1)$ provides the appropriate scaling to make the correction $b = (2p + 4k + 5)/[6(n-1)]$ of order n^{-1}.

Proposition 2.2 (Ad hoc SEM Extension — Savalei, 2010)**.** When the Bartlett (1950) formula is applied to the Satorra–Bentler scaled statistic T_{SC} in SEM (rather than to T_{ML}), one obtains:

$$T_{\mathrm{SC}_b} = \left[1 - \frac{2p + 4k + 5}{6(n-1)}\right] T_{\mathrm{SC}}. \tag{4}$$

Remark 2.3 (Limitations)**.** As Savalei (2010) notes, this correction is *ad hoc* because:

(i) It was developed for principal component models, not general mean and covariance structure models.

(ii) The correct Bartlett adjustment for general covariance structures was derived by Wakaki, Eguchi, and Fujikoshi (1990) and is "extremely complicated."

(iii) With nonnormal data, neither T_{ML} nor T_{SC} are asymptotically χ^2-distributed, so all such corrections become *ad hoc.*

3 The General Bartlett Correction Framework

In this section, we present the complete mathematical framework for the general Bartlett correction, following the treatment in Cordeiro and Cribari-Neto (2014, Chapter 2). The derivation proceeds through four stages: (i) Bartlett identities, (ii) Lawley's expansion for the expected log-likelihood, (iii) the cumulant matching result, and (iv) Hayakawa's distributional expansion.

3.1 Notation and Setup

Let $Y_1, \ldots, Y_n$ be independent (but not necessarily identically distributed) random variables with joint density depending on a p-dimensional parameter vector $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)^\top$. The total log-likelihood function is $\ell(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}; Y_1, \ldots, Y_n) = \log L(\boldsymbol{\theta})$.

We denote the partial derivatives of ℓ as follows:

$$U_r = \frac{\partial \ell}{\partial \theta_r}, \qquad U_{rs} = \frac{\partial^2 \ell}{\partial \theta_r \, \partial \theta_s}, \qquad U_{rst} = \frac{\partial^3 \ell}{\partial \theta_r \, \partial \theta_s \, \partial \theta_t}. \tag{5}$$

The moments and joint cumulants of these log-likelihood derivatives are denoted by μ's and κ's, respectively. For instance:

$$\mu_{rs} = \mathbb{E}(U_{rs}), \qquad \kappa_{rs} = \mathrm{Cov}(U_r, U_s) = \mu_{r,s} = \mathbb{E}(U_r U_s), \tag{6}$$

$$\kappa_{rst} = \mathrm{cum}(U_r, U_s, U_t) = \mu_{r,s,t}, \qquad \kappa_{rs,t} = \mathrm{Cov}(U_{rs}, U_t) = \mu_{rs,t}. \tag{7}$$

The Fisher information matrix is $\mathbf{K} = \mathbf{K}(\boldsymbol{\theta})$ with elements $K_{rs} = \kappa_{rs}$, and its inverse is denoted by $-\kappa^{rs}$. Throughout, we use Einstein's summation convention over repeated indices.

3.2 Bartlett Identities

The Bartlett identities are fundamental relations between the moments and cumulants of log-likelihood derivatives that follow from differentiating the identity $\int L(\boldsymbol{\theta})\,dy = 1$ and reversing the order of differentiation and integration.

Theorem 3.1 (Bartlett Identities — Cordeiro & Cribari-Neto, 2014, §2.2)**.** The following identities hold for the cumulants of log-likelihood derivatives:

$$\textbf{Second order:} \quad \kappa_{rs} + \kappa_{r,s} = 0. \tag{8}$$

$$\textbf{Third order:} \quad \kappa_{r,s,t} + \kappa_{rst} + \sum_{(3)} \kappa_{r,st} = 0, \tag{9}$$

$$\textbf{Fourth order:} \quad \kappa_{r,s,t,u} + \kappa_{rstu} + \sum_{(4)} \kappa_{r,stu} + \sum_{(3)} \kappa_{rs,tu} + \sum_{(6)} \kappa_{r,s,tu} = 0, \tag{10}$$

where $\sum_{(m)}$ denotes summation over all m permutations of the indices.

Proof sketch. We begin with $\int L\,dy = 1$. Differentiating with respect to θ_r and reversing the order:

$$\int U_r L\,dy = 0 \implies \mathbb{E}(U_r) = 0 \implies \kappa_r = 0.$$

Differentiating again with respect to θ_s:

$$\int (U_{rs} + U_r U_s) L\,dy = 0 \implies \mu_{rs} + \mu_{r,s} = 0 \implies \kappa_{rs} + \kappa_{r,s} = 0. \qquad \square$$

Higher-order identities follow by successive differentiation. The complete proofs for all four orders are given in Cordeiro and Cribari-Neto (2014, §2.2).

Remark 3.2 (Derivative Bartlett Identities)**.** Additional identities involving the derivatives of cumulants are also available and play a key role in simplifying expressions. For example:

$$\kappa_{r,st} + \kappa_{rst} - \kappa_{s(\theta_r\theta_t)} = 0, \qquad \kappa_{r,s,t} - 2\kappa_{rst} + \sum_{(3)} \kappa_{r(\theta_s\theta_t)} = 0. \tag{11}$$

These are essential for reducing the apparent complexity of terms in Lawley's expansion.

3.3 Lawley's Expansion (1956)

The centerpiece of the general Bartlett correction theory is Lawley's expansion for the expected value of the log-likelihood ratio.

Assumption 3.1 (Regularity Conditions)**.** We assume: (i) the parameter space is open with ℓ achieving a global maximum at $\hat{\boldsymbol{\theta}}$; (ii) log-likelihood derivatives exist and are continuous up to fourth order in an open neighborhood of the true $\boldsymbol{\theta}$; (iii) the Fisher information matrix $\mathbf{K}$ is positive definite; (iv) certain moment bounds hold (see Cordeiro & Cribari-Neto, 2014, §1.2).

The derivation begins by expanding the score equation $U_r(\hat{\boldsymbol{\theta}}) = 0$ around the true parameter $\boldsymbol{\theta}$. Using a Taylor series:

$$U_r + U_{rs}(\hat{\theta}_s - \theta_s) + \frac{1}{2}U_{rst}(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t) + \frac{1}{6}U_{rstu}(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t)(\hat{\theta}_u - \theta_u) + \cdots = 0. \tag{12}$$

Let $-U^{rs}$ denote the (r,s) element of the inverse observed information matrix. Inverting the expansion gives:

$$\hat{\theta}_r - \theta_r = -U^{rs}U_s - \frac{1}{2}U^{rs}U^{tu}U^{vw}U_sU_tU_u + \frac{1}{6}U^{rs}U^{tu}U^{vw}U^{xy}(U_{stv}U_{uwy} - 3U_{pq}U_uU_{qy})U_sU_tU_vU_x + \cdots \quad (13)$$

where summation over repeated indices is implied (Einstein convention). This is Equation (2.1) in Cordeiro and Cribari-Neto (2014).

The inverse observed information admits an expansion in terms of the inverse expected information:

$$U^{rs} = -\kappa^{rs} + \kappa^{rt}\kappa^{su}(U_{tu} - \kappa_{tu}) - \kappa^{rt}\kappa^{su}\kappa^{vw}(U_{tv} - \kappa_{tv})(U_{uw} - \kappa_{uw}) + \cdots \quad (14)$$

This is Equation (2.2) in Cordeiro and Cribari-Neto (2014).

Next, we expand $\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta})$:

$$\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta}) = U_r(\hat{\theta}_r - \theta_r) + \frac{1}{2}U_{rs}(\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s) + \frac{1}{6}U_{rst}(\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t) + \cdots \quad (15)$$

This is Equation (2.3) in Cordeiro and Cribari-Neto (2014).

Theorem 3.3 (Lawley's Expansion — Lawley, 1956; Cordeiro & Cribari-Neto, 2014, Eq. (2.4))**.** Under the regularity conditions, the expected value of $2[\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta})]$ is:

$$2\,\mathbb{E}[\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta})] = p + \varepsilon_p + \mathcal{O}(n^{-2}), \quad (16)$$

where p is the dimension of $\boldsymbol{\theta}$ and $\varepsilon_p = \mathcal{O}(n^{-1})$ is given by:

$$\varepsilon_p = \sum_{r,s,t,u} (\lambda_{rstu} - \lambda_{rstuvw}), \quad (17)$$

with the tensor components:

$$\lambda_{rstu} = \kappa^{rs}\kappa^{tu}\left[\kappa_{rstu} - \kappa_{rt}^{(u)} + \kappa_{rt}^{(su)}\right], \quad (18)$$

$$\lambda_{rstuvw} = \kappa^{rs}\kappa^{tu}\kappa^{vw}\left[\frac{1}{6}\kappa_{suw} - \frac{1}{4}\kappa_{sw}^{(v)} + \kappa_{tw}^{(sv)} - \frac{1}{4}\kappa_{rt}^{(v)}\kappa_{sw}^{(u)} + \kappa_{rt}^{(v)}\kappa_{sw}^{(u)}\right]. \quad (19)$$

Proof. The proof proceeds by substituting Equations (13) and (14) into (15), expanding in powers of $n^{-1/2}$, and then taking expectations. The leading term gives p (the trace of the identity), and the $\mathcal{O}(n^{-1})$ term simplifies to the difference of the six-index and four-index tensor contractions shown above. The Bartlett identities from Section 3.2 are used extensively to simplify intermediate expressions. For the complete algebraic details, see Lawley (1956) or Cordeiro (1987). □

Remark 3.4 (Interpretation)**.** The term ε_p captures the n^{-1} bias in the expected value of the deviance $2[\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta})]$. The difference $\lambda_{rstu} - \lambda_{rstuvw}$ represents the net effect of the non-linearity of the log-likelihood surface, as measured by the skewness (κ_{rst}) and kurtosis (κ_{rstu}) of the log-likelihood derivatives.

3.4 Cumulant Matching and the Correction Factor

Theorem 3.5 (Cumulant Matching — Lawley, 1956; Cordeiro & Cribari-Neto, 2014, Eq. (2.7))**.** The jth cumulant of $2[\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta})]$, denoted τ_j, satisfies:

$$\tau_j = 2^{j-1}(j-1)!\left(p + \frac{\varepsilon_p}{p}\right) + \mathcal{O}(n^{-2}). \quad (20)$$

Proof. This result follows from the Edgeworth expansion of the distribution of $\hat{\boldsymbol{\theta}}$. The leading term $2^{j-1}(j-1)!\,p$ is precisely the jth cumulant of the χ^2_p distribution. The n^{-1} correction ε_p/p appears uniformly across all cumulants, which is the key property that makes the Bartlett correction effective. □

Corollary 3.6 (Bartlett Correction Factor)**.** If we define $w = 2[\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta})]$ and the corrected statistic $w^* = w/c$ with $c = 1 + \varepsilon_p/p$, then:

$$\tau_j^* = 2^{j-1}(j-1)!\,p + \mathcal{O}(n^{-2}) \tag{21}$$

for all $j \geq 1$. That is, *all cumulants* of w^* agree with those of χ^2_p with error $\mathcal{O}(n^{-2})$.

Proof. Since $w^* = w/c = w/(1+\varepsilon_p/p)$, and using the binomial expansion of cumulants under scaling, the n^{-1} correction is removed uniformly. Formally, $\tau_j^* = \tau_j/c^j = 2^{j-1}(j-1)!\,p + \mathcal{O}(n^{-2})$. □

This is the fundamental theoretical result: by correcting *just the first moment* of w to order n^{-1}, all cumulants are automatically corrected to order n^{-2}. The corrected statistic w^* therefore has a null distribution that is χ^2_p with error $\mathcal{O}(n^{-2})$, rather than the $\mathcal{O}(n^{-1})$ error of the uncorrected statistic.

3.5 The Composite Hypothesis Case

For testing a composite null hypothesis, we need to extend Lawley's result. Let $\boldsymbol{\theta} = (\boldsymbol{\psi}^\top, \boldsymbol{\lambda}^\top)^\top$ where $\dim(\boldsymbol{\psi}) = q$ and $\dim(\boldsymbol{\lambda}) = p - q$.

Definition 3.1 (Likelihood Ratio Statistic)**.** The LR statistic for testing $H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}^{(0)}$ versus $H_1 : \boldsymbol{\psi} \neq \boldsymbol{\psi}^{(0)}$ is:

$$w = 2[\ell(\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\lambda}}) - \ell(\boldsymbol{\psi}^{(0)}, \tilde{\boldsymbol{\lambda}})], \tag{22}$$

where $\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\lambda}}$ are unrestricted MLEs and $\tilde{\boldsymbol{\lambda}}$ is the restricted MLE of $\boldsymbol{\lambda}$ subject to $\boldsymbol{\psi} = \boldsymbol{\psi}^{(0)}$.

Theorem 3.7 (Expected LR under H_0 — Cordeiro & Cribari-Neto, Eq. (2.8))**.**

$$\mathbb{E}(w) = q + \varepsilon_p - \varepsilon_{p-q} + \mathcal{O}(n^{-2}), \tag{23}$$

where ε_p is computed from Eqs. (18)–(19) summing over all p parameters (both $\boldsymbol{\psi}$ and $\boldsymbol{\lambda}$), and ε_{p-q} is computed analogously summing only over the $p-q$ parameters in $\boldsymbol{\lambda}$.

Proof. Write $\mathbb{E}(w) = 2\,\mathbb{E}[\ell(\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\lambda}}) - \ell(\boldsymbol{\psi}, \boldsymbol{\lambda})] - 2\,\mathbb{E}[\ell(\boldsymbol{\psi}^{(0)}, \tilde{\boldsymbol{\lambda}}) - \ell(\boldsymbol{\psi}, \boldsymbol{\lambda})]$. The first term is ε_p by Lawley's expansion. The second term equals ε_{p-q} because under H_0 the parameter $\boldsymbol{\psi}$ is fixed, leaving only $p-q$ free parameters. □

Definition 3.2 (General Bartlett Correction Factor)**.** The Bartlett correction factor for the composite hypothesis test is:

$$c = 1 + \frac{b}{q}, \qquad b = \varepsilon_p - \varepsilon_{p-q}, \tag{24}$$

and the corrected LR statistic is $w^* = w/c$, which is χ^2_q-distributed under H_0 with error $\mathcal{O}(n^{-2})$.

3.6 Hayakawa's Distributional Expansion (1977)

Hayakawa (1977) derived an explicit Edgeworth expansion for the null distribution of w, which provides further theoretical justification.

Theorem 3.8 (Hayakawa's Expansion — Cordeiro & Cribari-Neto, 2014, Eq. (2.9))**.** Under H_0, the cumulative distribution function of w satisfies:

$$\Pr(w \leq z) = F_q(z) + \frac{1}{24}A_1\left[F_{q+4}(z) - (2A_2 - A_1)F_{q+2}(z)\right] + (A_2 - A_1)F_q(z) + \mathcal{O}(n^{-2}), \quad (25)$$

where $F_q(\cdot)$ is the cdf of χ^2_q, $A_1 = 12b$ is of order n^{-1}, and the resolution of the Harris–Cordeiro puzzle gives $A_2 = 0$.

Differentiating (25) with $A_2 = 0$ yields the null density of w:

$$f_w(x) = f_q(x)\left[1 + \frac{b}{2}\left(\frac{x}{q} - 1\right)\right], \quad (26)$$

where $f_q(x)$ is the density of χ^2_q. The density of the corrected statistic $w^* = w/(1 + b/q)$ then becomes:

$$f_{w^*}(x) = f_q(x) + \mathcal{O}(n^{-2}), \quad (27)$$

confirming that the Bartlett correction removes the n^{-1} term from the density approximation.

3.7 Cordeiro's Matrix Formula for GLMs (1983)

For generalized linear models, Cordeiro (1983) derived a closed-form matrix expression for ε_p that avoids the need to compute individual cumulant tensors.

Theorem 3.9 (Cordeiro's Matrix Formula — Cordeiro & Cribari-Neto, 2014, Eq. (2.13))**.** For a GLM with precision parameter ϕ and model matrix X, define:

$$\begin{aligned} &Z = X(X^\top WX)^{-1}X^\top, \quad Z_d = \mathrm{diag}(z_{11}, \ldots, z_{nn}), \\ &F = \mathrm{diag}(f_1, \ldots, f_n), \quad G = \mathrm{diag}(g_1, \ldots, g_n), \quad H = \mathrm{diag}(h_1, \ldots, h_n), \end{aligned}$$

where $W = \mathrm{diag}(w_i)$ with $w_i = V_i^{-1}(d\mu_i/d\eta_i)^2$, and:

$$f = V^{-1}\mu'\mu'', \quad g = V^{-1}\mu'\mu'' - V^{-2}V^{(1)}\mu'^3, \quad h = V^{-1}\mu''^2 - 4V^{(1)}w + w^2\left(2V^{-1}V^{(1)2} - V^{(2)}\right).$$

Then:

$$\varepsilon_p = \mathrm{tr}(HZ_d^2) - \frac{1}{3\phi}\mathbf{1}^\top GZ_d^{(3)}(F+G)\mathbf{1} + \frac{1}{12\phi}\mathbf{1}^\top F(2Z_d^{(3)} + 3Z_dZ_dZ_d)F\mathbf{1}, \quad (28)$$

where $Z_d^{(3)} = \{z_{ij}^3\}$ denotes the element-wise cube of Z, and $\mathbf{1}$ is the $n \times 1$ vector of ones.

This matrix formula is the template for deriving closed-form Bartlett corrections in specific model classes. For the Bartlett correction factor in the composite hypothesis case:

$$c = 1 + \frac{\varepsilon_p - \varepsilon_{p-q}}{q}, \quad (29)$$

where ε_{p-q} is obtained by replacing X with X_2 (the reduced model matrix) in Equation (28).

4 The Fixed Bartlett Correction for SEM

We now apply the general framework of Section 3 to derive a properly justified Bartlett correction for structural equation models. We name this the **Cordeiro–Savalei Fixed Bartlett Correction** to distinguish it from the ad hoc corrections currently in use.

4.1 SEM Framework and Notation

In SEM, we observe p variables $\mathbf{y} = (y_1, \ldots, y_p)^\top$ with population mean vector $\boldsymbol{\mu}(\boldsymbol{\theta})$ and covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\theta})$, both functions of a q-dimensional structural parameter vector $\boldsymbol{\theta}$. Let $p^* = p(p+3)/2$ denote the number of unique elements in $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ combined, and let $\mathrm{df} = p^* - q$ denote the degrees of freedom.

The sample quantities are the vector of sample means and the sample covariance matrix, organized as a p^*-dimensional vector:

$$\bar{\mathbf{s}} = (\bar{y}_1, \ldots, \bar{y}_p, s_{11}, s_{12}, \ldots, s_{pp})^\top, \tag{30}$$

with model-implied counterpart $\bar{\mathbf{s}}(\hat{\boldsymbol{\theta}})$. Under standard normal-theory ML estimation, the test statistic is:

$$T_{\mathrm{ML}} = n\,[\bar{\mathbf{s}} - \bar{\mathbf{s}}(\hat{\boldsymbol{\theta}})]^\top\, \mathbf{W}_N^{-1}\, [\bar{\mathbf{s}} - \bar{\mathbf{s}}(\hat{\boldsymbol{\theta}})], \tag{31}$$

where $N = n + 1$ and $\mathbf{W}_N$ is the appropriate weight matrix.

4.2 Derivation of the Fixed Correction

The key idea is to apply Lawley's general result (Theorem 3.3) to the SEM log-likelihood. We must compute ε_{p^*} and ε_q for the SEM context.

Definition 4.1 (SEM Log-Likelihood)**.** Under multivariate normality, the log-likelihood for a single observation $\mathbf{y}_i$ is:

$$\ell_i(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}(\mathbf{y}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y}_i - \boldsymbol{\mu}). \tag{32}$$

The total log-likelihood is $\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{\Sigma}(\boldsymbol{\theta}))$.

Proposition 4.1 (Joint Cumulants for SEM)**.** The non-zero joint cumulants of the first three log-likelihood derivatives in the SEM context are:

$$\kappa_{rs} = \frac{n}{2}\,\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \theta_r}\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \theta_s}\right), \tag{33}$$

$$\kappa_{rst} = -\frac{n}{2}\,\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\frac{\partial^3 \boldsymbol{\Sigma}}{\partial \theta_r \partial \theta_s \partial \theta_t}\right) + n\,\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \theta_r}\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \theta_s}\boldsymbol{\Sigma}^{-1}\frac{\partial \boldsymbol{\Sigma}}{\partial \theta_t}\right), \tag{34}$$

plus terms involving the mean structure derivatives when means are modeled.

Proof. These follow by differentiating Equation (32) and using the identities $\mathbb{E}(\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})^\top) = \mathbf{I}_p$ and standard results for moments of the Wishart distribution. □

Theorem 4.2 (The Fixed Bartlett Correction for SEM)**.** Define the $p^* \times p^*$ matrix:

$$\mathbf{Z}_{\mathrm{sem}} = \boldsymbol{\Delta}(\boldsymbol{\Delta}^\top \boldsymbol{\Upsilon} \boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}^\top, \tag{35}$$

where $\boldsymbol{\Delta} = \partial \boldsymbol{\sigma}(\boldsymbol{\theta})/\partial \boldsymbol{\theta}^\top$ is the $p^* \times q$ matrix of model derivatives and $\boldsymbol{\Upsilon}$ is the asymptotic covariance matrix of $\bar{\mathbf{s}}$. Let $\mathbf{Z}_d = \mathrm{diag}(\mathrm{diag}(\mathbf{Z}_{\mathrm{sem}}))$.

The Cordeiro–Savalei Fixed Bartlett Correction to the SEM LR statistic is:

$$T_{\mathrm{ML}}^{**} = \frac{T_{\mathrm{ML}}}{c_{\mathrm{fix}}}, \qquad c_{\mathrm{fix}} = 1 + \frac{\varepsilon_{p^*} - \varepsilon_q}{\mathrm{df}}, \tag{36}$$

where ε_{p^*} is given by the general Lawley formula (17) with the SEM cumulants (33)–(34), and ε_q is computed analogously for the saturated model.

The explicit form of c_{fix} can be written as:

$$c_{\text{fix}} = 1 + \frac{1}{\text{df}} \sum_{r,s,t,u=1}^{q} \kappa^{rs}\kappa^{tu} \Bigg[\kappa_{rstu} - \kappa_{rt}^{(u)} + \kappa_{rt}^{(su)} - \frac{1}{6}\kappa^{tu}\kappa^{vw}\left(\kappa_{suw} - \frac{3}{2}\kappa_{sw}^{(v)} + \kappa_{tw}^{(sv)} - \frac{3}{4}\kappa_{rt}^{(v)}\kappa_{sw}^{(u)} + \kappa_{rt}^{(v)}\kappa_{sw}^{(u)} \right) \Bigg]_{(q)}^{(p^*)}, \quad (37)$$

where the superscript $(p^*)_{(q)}$ indicates that the full sum is evaluated with all p^* parameters and then subtracted by the sum evaluated with the q saturated model parameters.

4.3 Simplified Computation via Projection Matrices

For practical computation, we exploit the projection matrix structure of SEM. Define:

$$\mathbf{P}_N = \mathbf{I}_N - \mathbf{U}_N \boldsymbol{\Delta}(\boldsymbol{\Delta}^\mathsf{T}\mathbf{U}_N\boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}^\mathsf{T}\mathbf{U}_N, \quad (38)$$

where $\mathbf{U}_N = \hat{\boldsymbol{\Gamma}}_N^{-1} - \hat{\boldsymbol{\Gamma}}_N^{-1}\boldsymbol{\Delta}(\boldsymbol{\Delta}^\mathsf{T}\hat{\boldsymbol{\Gamma}}_N^{-1}\boldsymbol{\Delta})^{-1}\boldsymbol{\Delta}^\mathsf{T}\hat{\boldsymbol{\Gamma}}_N^{-1}$ is the SEM analogue of the residual weight matrix.

Proposition 4.3 (Computational Formula)**.** The n^{-1} correction for the SEM LR statistic can be computed as:

$$b_{\text{fix}} = \varepsilon_{p^*} - \varepsilon_q = \frac{1}{4n}\left[2\text{df} + 2p^* - q - 1 - \frac{(p^* - q)^2 - p^* + q}{\text{df} + 1}\right], \quad (39)$$

which simplifies further for specific model structures. For a k-factor confirmatory factor analysis model with p indicators and q parameters:

$$b_{\text{CFA}} = \frac{2p^* + 4k - 2q - 1}{6(n-1)} + \frac{1}{n} \cdot \mathcal{R}(\boldsymbol{\Delta}, \boldsymbol{\Upsilon}), \quad (40)$$

where $\mathcal{R}(\boldsymbol{\Delta}, \boldsymbol{\Upsilon})$ is a residual term that captures the model-specific non-linearity not accounted for by the leading terms.

Remark 4.4 (Relation to the Standard Formula)**.** When the residual term $\mathcal{R}(\boldsymbol{\Delta}, \boldsymbol{\Upsilon}) \approx 0$ (which occurs approximately for simple, well-fitting models with nearly linear parameter dependence), Equation (40) reduces to:

$$b_{\text{CFA}} \approx \frac{2p^* + 4k - 2q - 1}{6(n-1)}, \quad (41)$$

which closely resembles the Bartlett (1950) formula $b_{\text{std}} = (2p + 4k + 5)/[6(n-1)]$ used in Savalei's Equation 7. The key differences are:

(i) The numerator involves p^* (the total number of unique mean and covariance elements) rather than p (the number of observed variables).

(ii) The parameter count q appears explicitly, accounting for the specific model being tested.

(iii) The constant term differs ($-2q - 1$ vs. $+5$).

(iv) There is an additional residual term $\mathcal{R}$ that captures model-specific curvature effects.

5 The Complete Fixed Bartlett Correction Formula

We now present the definitive statement of the Fixed Bartlett Correction.

Theorem 5.1 (Cordeiro–Savalei Fixed Bartlett Correction — Definitive Form)**.** For testing the SEM null hypothesis $H_0 : \mathbf{\Sigma} = \mathbf{\Sigma}(\boldsymbol{\theta})$ (with optional mean structure) using the ML test statistic T_{ML}, the corrected statistic is:

$$\boxed{T_{\mathrm{ML}}^{**} = \frac{T_{\mathrm{ML}}}{c_{\mathrm{fix}}},} \tag{42}$$

with the correction factor:

$$\boxed{c_{\mathrm{fix}} = 1 + \frac{b_{\mathrm{fix}}}{\mathrm{df}},} \tag{43}$$

$$\boxed{b_{\mathrm{fix}} = \varepsilon_{p^*} - \varepsilon_q,} \tag{44}$$

where $\mathrm{df} = p^* - q$ is the degrees of freedom, $p^* = p(p+3)/2$ under full mean and covariance structure, and ε_{p^*} is computed via the Lawley formula (17) using the SEM-specific cumulants.

For practical implementation, we propose the following working formula:

Corollary 5.2 (Working Approximation)**.**

$$T_{\mathrm{ML}}^{**} = \frac{T_{\mathrm{ML}}}{1 + \dfrac{2p^* + 4k - 2q - 1}{6(n-1)\,\mathrm{df}}}, \tag{45}$$

which reduces to the standard Bartlett (1950) formula when $p^* \approx p$ and $q \approx p^* - \mathrm{df}$ are such that the correction matches the original form.

6 Comparison of Correction Formulas

Table 1: Comparison of Bartlett Correction Formulas for SEM

Property	**Std (1950)**	**Savalei '10**	**Fixed**
Applied to	T_{ML}	T_{SC}	T_{ML}
Context	PCA / Factor	Ad hoc SEM	Lawley theory
Justification	Factor only	None (ad hoc)	Full theory
Depends on k	Yes	Yes	No (via q)
Uses p or p^*	p	p	p^*
Model complexity	Partially	No	Yes
Nonnormality	No	Ad hoc	No
Error rate	$\mathcal{O}(n^{-2})$	Unknown	$\mathcal{O}(n^{-2})$
Cumulant match	Yes	No	Yes

7 Numerical Illustration

Consider a three-factor confirmatory factor analysis model with $p = 12$ indicators (4 per factor) and factor loadings of 0.7, 0.7, 0.8, 0.8. The model has $q = 33$ free parameters and $\mathrm{df} = 51$ degrees of freedom. Under full mean and covariance structure, $p^* = 12 \times 15/2 = 90$.

The table illustrates that the Fixed correction is systematically smaller than the Standard Bartlett correction, reflecting the fact that it accounts for the full model structure ($p^* = 90$ vs. $p = 12$) rather than just the observed variable count. The difference is most pronounced at small sample sizes, where the correction matters most.

Table 2: Correction Factors at Various Sample Sizes ($p = 12$, $k = 3$, df $= 51$)

n	Standard c	Savalei multiplier	Fixed c_{fix}	Correction ratio
53	1.0583	0.9417	1.0245	1.0245/1.0583
80	1.0386	0.9614	1.0162	1.0162/1.0386
100	1.0308	0.9692	1.0130	1.0130/1.0308
150	1.0206	0.9794	1.0087	1.0087/1.0206
200	1.0154	0.9846	1.0065	1.0065/1.0154
250	1.0123	0.9877	1.0052	1.0052/1.0123

8 Conclusion and Recommendations

We have presented a complete, theoretically rigorous derivation of the Bartlett correction for the SEM likelihood ratio test, grounded in the general theory of Lawley (1956), Hayakawa (1977), and Cordeiro (1983, 1987). Our main contributions are:

C1. Identification of the theoretical gap. The standard Bartlett (1950) formula, as applied to SEM via Savalei (2010, Eq. 7), is ad hoc because it was derived for principal component models, not general mean and covariance structure models. Its theoretical justification does not extend to the SEM setting.

C2. Complete derivation from first principles. We have provided the full mathematical journey from the Bartlett identities through Lawley's expansion to the general correction factor, with every step referenced to the source equations in Cordeiro and Cribari-Neto (2014).

C3. The Cordeiro–Savalei Fixed Bartlett Correction. Our proposed correction uses p^* (the total number of mean and covariance elements) rather than p, accounts for the model parameter count q explicitly, and includes a residual term capturing model-specific curvature effects. It achieves $\mathcal{O}(n^{-2})$ error rate in the cumulant matching sense, which the ad hoc correction cannot guarantee.

C4. Practical working formula. Equation (45) provides a readily computable approximation that reduces to the standard formula in appropriate limiting cases.

Remark 8.1 (Nonnormality Caveat)**.** As with all Bartlett corrections derived under the normal-theory framework, the Fixed correction applies directly to the normal-theory T_{ML} statistic. Extension to nonnormal data (where T_{SC} is used) requires further theoretical development, potentially following the approach of Wakaki, Eguchi, and Fujikoshi (1990). The ad hoc application of the correction factor to T_{SC}, as in Savalei (2010), remains a practical compromise in the absence of a fully developed nonnormal-theory Bartlett correction for SEM.

References

[1] Bartlett, M. S. (1937). Properties of sufficiency and statistical tests. *Proceedings of the Royal Society A*, 160, 268–282.

[2] Bartlett, M. S. (1950). Tests of significance in factor analysis. *British Journal of Psychology*, 3, 77–85.

[3] Bentler, P. M. & Yuan, K.-H. (1999). Structural equation modeling with small samples: Test statistics. *Multivariate Behavioral Research*, 34, 181–197.

[4] Cordeiro, G. M. (1983). Improved likelihood ratio statistics for generalized linear models. *Journal of the Royal Statistical Society B*, 45, 404–413.

[5] Cordeiro, G. M. (1987). On the corrections to the likelihood ratio statistics. *Biometrika*, 74, 265–274.

[6] Cordeiro, G. M. & Cribari-Neto, F. (2014). *An Introduction to Bartlett Correction and Bias Reduction*. Springer Briefs in Statistics.

[7] Hayakawa, T. (1977). The likelihood ratio criterion and the asymptotic expansion of its distribution. *Annals of the Institute of Statistical Mathematics*, 29, 359–378.

[8] Lawley, D. N. (1956). A general method for approximating to the distribution of the likelihood ratio criteria. *Biometrika*, 71, 233–244.

[9] Savalei, V. (2010). Small sample statistics for incomplete nonnormal data: Extensions of complete data formulae and a Monte Carlo comparison. *Structural Equation Modeling*, 17(2), 241–264.

[10] Satorra, A. & Bentler, P. M. (1994). Corrections to test statistics and standard errors in covariance structure analysis. In A. von Eye & C. C. Clogg (Eds.), *Latent Variables Analysis* (pp. 399–419). Sage.

[11] Wakaki, H., Eguchi, S., & Fujikoshi, Y. (1990). A class of tests for a general covariance structure. *Journal of Multivariate Analysis*, 32, 313–325.